

Deployment readiness checklist

Module 6 - Lesson 3

Overview

Moving a model to production is engineering, not just science. A deployment checklist covers the API, monitoring, data pipeline, and rollback plan.

Key Concepts

- Expose the model behind a versioned API with input validation.
- Set up monitoring: prediction latency, request volume, and error rate.
- Detect data drift and prediction drift after deployment.
- Automate retraining on fresh data and validate before replacing the model.
- Plan rollback: keep the previous model version available and switchable.
- Document the runbook: what to do when metrics go wrong.

Example

```
Deployment checklist:  
- Model versioned and reproducible (code + data + config)  
- API validates input and returns errors clearly  
- Logging for latency, errors, and drift  
- Retraining pipeline scheduled and tested  
- Rollback path tested  
- Runbook written and owners assigned
```

Summary

Production ML is about reliability, not just accuracy. Versioned models, monitoring for drift, automated retraining, and a tested rollback path keep deployments safe.

Practice Checklist

- Write a deployment checklist tailored to your model.
- Add drift monitoring to the serving endpoint plan.
- Document the rollback procedure step by step.